Applications of binomial distributions

1 A fair six-sided die is rolled 5 times. Find the probability of rolling:

a Exactly 3 ones

b Less than 3 ones

c No more than 3 ones

2 Adam has product tested the name of his new product and has found that 61% of people like it. Assuming his product test results are correct, if he randomly asks 50 people if they like the name of his new product, find the probability that exactly 24 people will say they like the name. Round your answer to two decimal places.

3 The founders of a dating app claim that 39% of messages are replied to. Assuming that this is correct, if Aaron sends 30 messages to different people, find the probability that he will get replies to exactly half of his messages. Round your answer to four decimal places.

4 72% of the fans attending a particular concert are male. If a security guard decides to randomly check exactly 10 fans' bags for prohibited items, find the probability that he checks exactly 5 females' bags. Round your answer to four decimal places.

5 64% of penguins at a particular zoo are female. If a zookeeper chooses 6 penguins at random for health checks, find the probability that exactly 3 of them will be female. Round your answer to four decimal places.

6 A gym runs an offer for one week's free membership. When they ran this offer previously, they found that 42% of people then became full-time members. If 90 people take up the offer, find the probability that exactly 54 people become full-time members based on previous promotions. Round your answer to four decimal places.

7 24% of the students in a soccer team have scored a goal this season. If 4 members of the soccer team are chosen at random:

a Find the probability that all four of them have scored a goal this season. Round your answer to six decimal places.

b Find the probability that the first two members scored a goal this season, and the second two did not. Round your answer to four decimal places.

c Find the probability that any two of the four members scored a goal this season, and the other two did not. Round your answer to four decimal places.

8 Census data shows that 30% of the population in a particular country have red hair. Find the probability that more than half of a random sample of 6 people have red hair. Round your answer correct to four decimal places.

9 Lisa has created a toy frog that can hop. She claims that 78% of the hops are over 40 cm. Assuming Lisa's claim is correct, if the toy frog hops 6 times in a row, find the probability that

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at least 4 hops are over $40\ \text{cm}$. Round your answer to four decimal places.

10 Records show that half of all households in a city have broadband. Find the probability that less than 45% of a sample of 100 random households have a broadband. Round your answer to four decimal places.

11 Deborah has a drinks factory. Bottles are automatically filled and then a label is placed on them. When they are placed into their boxes, there is a 86% chance that each bottle is placed with the label facing the front of the box. There are 12 bottles placed into each box. Find the probability that none of the bottles will have the label facing the front of the box. Give your answer to two significant figures in scientific notation.

12 A boy's favourite toy is a plastic dinosaur. In a box of 20 toys, there are 8 such dinosaurs. Without looking, the boy takes a toy and plays with it for awhile and then puts it back. He does this 9 times.
The number of toy dinosaurs, x, the boy plays with can be modelled by a discrete probability distribution function of the form:

$$P(x) = {}^{9}C_{x}\,(0.4)^{x}\,(0.6)^{9-x}\ ;\ x = 0, 1, 2, \ldots, 9$$

a State the type of distribution the function characterises and describe what the value of 0.4 in the function represents.

b Find each of the following to three decimal places:

i The probability that the boy plays with exactly 2 toy dinosaurs.

ii The probability that the boy plays with at most 1 toy dinosaur.

iii The probability that the boy plays with at least 2 toy dinosaurs given that he played with at most 8 toy dinosaurs.

13 A particular nationwide numeracy test has a failure rate of 30%. If you randomly selected 100 students from across the country to do the test, how many would you expect to pass?

14 The chance of rain on any day in November is 25%.

a On how many days do you expect rain?

b Describe why modelling the number of rainy days in a given month, from previous weather data, is not well suited to a binomial distribution.

15 The probability that a tomato seed germinates is 0.75. Find n, the number of seeds you should plant if you want to grow 6 tomato plants.

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16 A science exam consisted of 48 multiple choice questions, each with 4 possible options. Ray guessed the answers to all of the questions at random. Let X be the number of questions Ray gets correct.

a Find $E(X)$.

b Find $\text{Var}(X)$.

c Find the standard deviation of X.

17 A science exam consisted of 75 multiple choice questions, each with 5 possible options. Bart guessed the answers to all the questions at random. Let X be the number of questions Bart gets correct.

a Find $E(X)$.

b Find $\text{Var}(X)$.

18 Consider the following binomial distributions:

- Let X be the number of questions answered correctly in a 40 question multiple choice test when all answers are guessed and each question has 5 possible responses.
- Let Y be the number of times a four results from rolling an 8-sided dice 64 times.

a Find the following, correct to two decimal places when necessary:

i Mean of X

ii Mean of Y

iii Standard deviation of X

iv Standard deviation of Y

b Is X or Y more likely to have a value between 6 and 10?

c Find $P(6 \leq X \leq 10)$ to two decimal places.

d Find $P(6 \leq Y \leq 10)$ to two decimal places.

19 A research study found that professional football players successfully kicked penalty goals 72% of the time. Frank kicked 49 penalty goals in his career. Let the number of penalty goals Frank missed be X.

a Find $E(X)$.

b Find the variance $\text{Var}(X)$ correct to one decimal place.

20 In a certain electorate, 12% of voters usually vote for the L.E.F.T party. In the upcoming election, there are 6000 people enrolled to vote.

a Find the number of people expected to vote for the L.E.F.T party.

b Find the standard deviation for the number of voters who do not vote for the L.E.F.T party. Round your answer to two decimal places.

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21 James has a cabbage patch that has been attacked by cabbage moth. Let X be the number of cabbages ruined, where X is a binomial distribution with:

- n, the number of cabbages James planted
- p, the probability of each being struck by the cabbage moth

a Find the value of n and p, given he expects to have 5 of his cabbages ruined with a standard deviation of 2.

b Given that more than 2 were ruined, find the probability that at most 4 were ruined. Round your answer to two decimal places.

22 A multiple choice test has 15 questions in total. Each question has 6 options, only one of which is correct. A student has not studied at all for the test and guesses the answers to all of the questions at random.
Find the following, rounding your answers to four decimal places:

a The probability that the student answers exactly 3 questions correctly.

b The probability that the student answers at most 4 questions correctly.

c The probability that the student answers between 2 and 6 questions inclusive correctly.

d The probability that they answered less than 6 correctly given that the student answered at least 2 questions correctly.

e The probability that the first 4 questions are answered correctly, and the rest are not.

23 In Western Australia, it has been shown that 40% of all voters are in favour of daylight saving. A sample of 15 voters are selected at random.
Find the following, rounding your answers to five decimal places:

a Find the probability that none of these voters are in favour of daylight saving.

b Find the probability that exactly 8 are in favour of daylight saving.

c Find the probability that exactly 5 are not in favour of daylight saving.

d Given that at least 2 are in favour, find the probability that at most 4 are in favour.

24 In the general population, it is estimated that 20% of people suffer with depression. A study group of 40 people is selected and tested for depression. Let X represent the number of people in the study group who are found to have depression.

a Find the number of people in the study group you expect to have depression.

b Find the exact standard deviation of X.

c Find the probability that the number of people with depression in the study group is within two standard deviations of the mean. Round your answer to two decimal places.

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25 Mohamad is currently applying for graduate job positions. For each application he submits, the probability that it gets short-listed and invited for an interview is 0.02.

a If he applies for 8 positions, find the probability that he will not get a single interview. Round your answer to two decimal places.

b If he applies for 8 positions, find the probability that he will get at least one interview. Round your answer to two decimal places.

c Write an expression for the probability that he will get at least one interview if he applies for n positions.

d Hence, find the minimum number of applications he will have to submit to ensure that the probability that he gets at least one interview is greater than 0.95.

e Mohamad has been applying for 8 positions every week over the last 12 weeks. Define the random variable Y, where Y is the number of weeks Mohamad had at least one interview. Complete the following, using the rounded probability found in part (b):

$$Y \sim B\left(\square, \square\right), \text{hence } P\left(Y = y\right) = \binom{\square}{\square} \times 0.15^{\square} \times 0.85^{\square}$$

f Find the probability that Mohamad had at least one interview per week, for at most 3 out of the 12 weeks. Round your answer to two decimal places.

26 A school choir has 30 members, and on a Monday morning, there is a 20% chance of each person not showing up for rehearsal.

a The choir will not rehearse unless at least $\frac{2}{3}$ of the members are present. Find the probability of the rehearsal being cancelled. Round your answer to four decimal places.

b The choir has 40 Monday morning rehearsals scheduled this year. How many rehearsals would you expect to be cancelled? Round your answer to the nearest whole number.

27 Maria buys a box of 8 glasses which is on sale because the box is expected to have around 3 broken glasses inside.

a Find the probability p that a glass is broken.

b Find the probability that at most half of the glasses are broken. Round your answer to two decimal places.

c At the store, there are 12 boxes in the sale, each containing 8 glasses. All of the boxes are expected to have about 3 broken glasses. Define the random variable X, where X is the number of boxes containing at most half of the glasses broken. Complete the following:

$$P\left(X = x\right) = \binom{\square}{\square} \times 0.86^{\square} \times 0.14^{\square}$$

d Hence, determine the probability that there are exactly 10 boxes on display where at most half of the glasses inside are broken. Round your answer to two decimal places.

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28 A restaurant finds that, on average, 6% of meals are sent back. On a particular night, there are 200 people served.

a Find the following, rounding your answers to four decimal places:

i The probability that exactly 6 meals are sent back.

ii The probability that within the first 100 customers, 10 or more meals are sent back.

iii The probability that within the first 100 customers, the 100th customer was the tenth customer to send their meal back that evening.

b If within the first 100 customers, ten people sent their meal back, should management be alerted to a possible problem in the kitchen? Explain your answer.

29 In a primary school, it is known that 27% of students have a serious allergy. 6 students visit the Health Nurse on a Monday.
Find the following, rounding your answers to four decimal places:

a The probability that the first three students to visit all have an allergic reaction.

b The probability that the first and the last student were the only students presenting with an allergic reaction.

c The probability that less than half of the students have an allergic reaction.

d The probability that at most 4 had an allergic reaction, given that at least 2 students had an allergic reaction.

30 In a batch of cheap envelopes, it is found that 1% have folds that won't stick down and are therefore defective. A pack of 30 of these envelopes are purchased. Assume that whether a particular envelope in a pack is defective is independent of whether any other envelope in the pack is defective.

a Find the probability that the first two envelopes taken from the packet are both defective.

b Find the probability that of the first two taken from the packet, only one is defective.

c Find the probability that exactly half of the envelopes in a packet are defective. Give your answer to two significant figures in scientific notation.

d Find the probability that less than 3 envelopes in a packet are defective. Round your answer to four decimal places.

e Given that more than 1 were defective, find the probability that at most 4 were defective. Round your answer to four decimal places.

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31 A basketballer has a 60% chance of successfully shooting any 3-point shot.

a If the basketballer takes 9 shots, find the probability that the first two go into the basket and the others don't. Round your answer to four decimal places.

b If the basketballer takes 9 shots, find the probability that at most 2 go into the basket. Round your answer to four decimal places.

c Find the minimum number of shots the basketball player needs to take so that the probability of him getting at least one in the basket is more than 99%.

32 A Facebook advertising campaign gets, on average, 200 Likes per 1000 users who see the advertisement.

a Find the probability that a user who sees the advertisement clicks 'Like'.

b A random sample of 30 users who have seen the advertisement are analysed for their response. Find the probability at least 2 clicked 'Like'. Round your answer to four decimal places.

c Given that in this sample more than 2 clicked 'Like', Find the probability that at most 4 clicked 'Like'. Round your answer to four decimal places.

d Find the smallest sample size that would lead to the probability of at least one user who has seen the advertisement clicking 'Like' being greater than 0.99.

33 In a particular suburb, crime rates are up, with 1 in every 4 households experiencing an attempted burglary. An insurance company randomly selects two samples, one of size 21 and one of size 24 from this suburb.

a Find the probability that both samples have exactly 10 households who have experienced an attempted burglary. Round your answer to five decimal places.

b Find the probability that the first sample has 8 households who have experienced an attempted burglary, while the second sample has 7 households. Round your answer to four decimal places.

c Find the probability that over the two samples, at least 44 households had experienced an attempted burglary. Give your answer to two significant figures in scientific notation.

34 At a particular doctor's surgery, 15% of patients come to hear about the results of a blood test. Two random samples of patients are taken: one of size 17 and one of size 27.

a Find the probability that both samples had exactly 5 patients who came to hear the results of a blood test. Round your answer to four decimal places.

b Find the probability that across both samples there were a total of less than 2 patients who came to hear the results for a blood test. Round your answer to four decimal places.

c Find the probability that across both samples there were exactly 3 patients who came to hear the results for a blood test, with more patients coming from the sample of size 17. Round your answer to four decimal places.

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35 A fair six-sided die is rolled 4 times.

a Find the probability of rolling a three the first 2 times and then rolling non-threes the last 2 times.

b Find the probability of rolling two threes and two non-threes in any order.

c 10 students are comparing their 4 rolls of the dice. Define the random variable X, where X is the number of students who rolled 2 threes. State the function for $P(X = x)$.

d Find the probability that less than 3 students rolled 2 threes. Round your answer to two decimal places.

e One of the students will continue to roll a die until she has seen 10 ones. Find the probability that this will take exactly 70 rolls of the die. Round your answer to two decimal places.

36 Yuri is playing a game in which he tosses a bunch of dice into the air and he wins if exactly 3 of the dice land on three.

a Write an expression to calculate the probability that Yuri will win if he tosses n dice.

b Hence, find the fewest number of dice he must toss to ensure that the probability that he wins is more than 15%.

37 Robyn does archery every Saturday. The probability that she hits the bullseye is 0.65 on each attempt. This Saturday Robyn has shots at the target 18 times.
Round answers to four significant figures where necessary.

a Find the probability that she will hit exactly 13 bullseyes.

b Find the probability that she will hit at most 10 bullseyes.

c Find the probability that she hits the bullseye on the first 8 attempts and none of the rest.

d Find the most likely number of bullseyes she will hit.

e Find the expected number of bullseyes she will hit.

f Robyn attends target practice for 10 consecutive Saturdays. Find the probability that on exactly half of these Saturdays, she hits the bullseye exactly 13 times.

g One Saturday Robyn decides to continue her target practice until she has hit the bullseye 15 times. Find the probability that she needs 24 shots at target to achieve her goal.

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